



CAREERSENSE ACADEMY
PROFESSIONAL LEARNING HANDBOOK

STATISTICS

for Data Professionals

From descriptive analysis to experiments, inference, regression, and evidence-based decisions

For data analysts • product analysts • BI professionals • data scientists • consultants • career switchers

A practical handbook for professionals who need to explain what the data means - not only calculate it.

CareerSense Professional Series • careersenseai.com/resources

How to Use This Handbook

Statistics becomes valuable when it helps a team make a better decision. This handbook is designed around that professional reality: define the question, choose the right statistical tool, check assumptions, quantify uncertainty, and communicate the result without overstating what the data can prove.

What each chapter contains

Each chapter follows a repeatable learning rhythm: business context, learning objectives, plain-English concepts, formulas only where they clarify reasoning, worked examples, interpretation guidance, common mistakes, revision notes, practice questions, and hands-on exercises.

How to practise

- Always write down the business question before choosing a test or metric.
- Calculate the statistic, then explain it in one sentence without jargon.
- Separate effect size from statistical significance.
- Check assumptions before trusting a model or test.
- Whenever possible, show uncertainty with an interval rather than only a point estimate.
- Treat charts as statistical arguments: label them clearly and show the relevant denominator.

PROFESSIONAL MINDSET

The goal is not to "use statistics." The goal is to reduce uncertainty enough to make a defensible decision, while being explicit about what remains unknown.

What You Will Learn

Chapter	Topic
01	Data Types, Grain, and Descriptive Statistics
02	Distributions, Shape, Center, and Spread
03	Probability and Random Variables
04	Sampling, Standard Error, and the Central Limit Theorem
05	Confidence Intervals and Estimation
06	Hypothesis Testing and p-values
07	Correlation and Linear Regression
08	Experiments and A/B Testing
09	Statistical Pitfalls, Bias, and Causality
10	Professional Case Study: CareerSense Job-Search Analytics

Closing the book: Whole-book revision • Professional glossary • 20-question quiz • Answer key

01

Data Types, Grain, and Descriptive Statistics

Before calculating an average, decide what one row represents. Many statistical errors are really grain errors hiding behind correct arithmetic.

Learning Objectives

- Distinguish categorical, ordinal, discrete, and continuous variables
- Define the grain of an analytical dataset
- Use count, proportion, mean, median, and mode appropriately
- Interpret range, variance, standard deviation, and interquartile range
- Choose descriptive metrics that match the business question

Main Concepts

A statistic summarizes data; it does not repair a badly defined dataset. Start with the observational unit - the thing represented by one row. A candidate-level table, an application-level table, and an interview-event table can all describe the same hiring journey but answer different questions.

Variable type	Example	Useful summaries
Nominal categorical	job_source	count, proportion, mode
Ordinal categorical	interview_rating 1-5	median, percentiles, distribution
Discrete numeric	applications_sent	mean, median, count
Continuous numeric	time_to_offer_days	mean, median, SD, IQR

Worked Example - time to first interview

Suppose six candidates took 4, 6, 7, 8, 9, and 30 days to receive a first interview. The mean is pulled upward by the 30-day case, while the median better represents the typical experience.

$$\text{Mean} \\ (4 + 6 + 7 + 8 + 9 + 30) / 6 = 10.7 \text{ days}$$

The average is mathematically correct but not especially representative.

Median

$$(7 + 8) / 2 = 7.5 \text{ days}$$

The median is resistant to the extreme 30-day value.

COMMON MISTAKE

Reporting an average without checking the distribution. If the data are skewed, the mean can describe the arithmetic center while misrepresenting the typical case.

Spread matters as much as center

Two teams can have the same average turnaround time but very different consistency. Standard deviation summarizes spread around the mean. The interquartile range (IQR) describes the middle 50% and is more robust to extreme values.

Interquartile Range

$$\text{IQR} = Q3 - Q1$$

Useful when the distribution is skewed or contains outliers.

Professional interpretation

A useful descriptive sentence includes the population, the metric, and the spread: "Among applicants who reached an interview, median time to first interview was 7.5 days, with the middle 50% between 5 and 9 days." That is far more informative than "average interview time was 10.7 days."

Quick Revision Notes

- Grain defines what one row represents.
- Categorical and numeric variables need different summaries.
- Mean is sensitive to extremes; median is resistant.
- Standard deviation summarizes spread around the mean.
- $\text{IQR} = Q3 - Q1$ and is robust to outliers.

Practice Questions

- Q1** Why should grain be defined before calculating metrics?
- Q2** When would median be more useful than mean?
- Q3** What does standard deviation communicate?
- Q4** What does IQR measure?
- Q5** Give one nominal and one ordinal variable from a hiring dataset.

Hands-on Exercises

1. Create a small dataset with one extreme value and compare mean vs median.
2. For a dataset you use at work, write down the grain in one sentence.
3. Choose three metrics and state which variable type each one summarizes.

02

Distributions, Shape, Center, and Spread

A single summary number can hide the most important feature of the data: its shape.

Learning Objectives

- Recognize symmetric, skewed, and multimodal distributions
- Understand percentiles and z-scores
- Interpret outliers in context
- Use robust statistics when appropriate
- Explain why the normal distribution is useful but not universal

Main Concepts

A distribution shows how values are spread across their possible range. Shape determines how much trust you should place in familiar summaries. Right-skewed data often appears in salaries, purchase values, and response times; left-skewed data can appear in bounded performance measures; multimodal data may indicate multiple populations mixed together.

Shape	What it often means	Professional response
Symmetric	center and tails are balanced	mean and SD may summarize well
Right-skewed	few very large values	report median/IQR; inspect tail
Multimodal	mixed segments or processes	segment before summarizing
Heavy-tailed	extremes occur more often	avoid normal assumptions blindly

Percentiles

The 90th percentile is the value at or below which 90% of observations fall. Percentiles are especially useful when stakeholders care about thresholds, service levels, or the experience of the slowest or fastest users.

z-score

$$z = (x - \text{mean}) / \text{standard deviation}$$

A z-score expresses how many standard deviations a value is from the mean.

BEST PRACTICE

Do not label an observation an "outlier" only because it is unusual. First ask whether it is a data error, a rare but valid event, or evidence of a different population.

The normal distribution

The normal distribution is a mathematical model with a symmetric bell shape. It is useful because many sampling distributions become approximately normal under broad conditions, even when the raw data are not. That does not mean every business variable should be forced into a normal model.

68-95 rule

About 68% within 1 SD; about 95% within 2 SD

This rule is approximate and applies to normally distributed values.

Worked Example - response time

If support response time has a mean of 12 minutes and SD of 3 minutes under a roughly normal process, a 15-minute response has $z = 1$. A 21-minute response has $z = 3$ and deserves investigation - but the z-score alone does not tell you whether the value is wrong.

Quick Revision Notes

- Distribution shape affects metric choice.
- Percentiles describe relative position.
- z-scores standardize distance from the mean.
- Normal assumptions should be justified, not automatic.
- Unusual values require context before removal.

Practice Questions

- Q1** What does right skew do to the mean relative to the median?
- Q2** What does the 95th percentile mean?
- Q3** Interpret $z = -2$ in plain English.
- Q4** Why can multimodality be analytically important?
- Q5** Why should valid extreme values not be deleted automatically?

Hands-on Exercises

4. Find a skewed metric in a dataset and report median plus IQR.
5. Calculate z-scores for five values.
6. Segment a mixed distribution by one business category and compare the shapes.

03

Probability and Random Variables

Probability gives data professionals a language for uncertainty before the outcome is known.

Learning Objectives

- Interpret probability as a long-run proportion or degree of uncertainty
- Apply complements, intersections, and conditional probability
- Distinguish independent from dependent events
- Understand discrete and continuous random variables
- Recognize Bernoulli and binomial settings

Main Concepts

Probability ranges from 0 to 1. In professional analytics, it often represents the chance of an event under a model: conversion, churn, failure, interview attendance, or defect occurrence. The useful question is usually conditional: what is the probability of an offer given that a candidate reached the final round?

Complement

$$P(\text{not } A) = 1 - P(A)$$

Conditional probability

$$P(A \text{ given } B) = P(A \text{ and } B) / P(B)$$

Concept	Example
Independent	two unrelated random samples
Dependent	interview pass and offer are linked
Bernoulli trial	candidate accepts offer: yes/no
Binomial count	number of acceptances among 20 offers

COMMON MISTAKE

Confusing $P(A \text{ given } B)$ with $P(B \text{ given } A)$. The direction of the condition matters, and reversing it can completely change the conclusion.

Worked Example - interview attendance

Assume 80% of scheduled candidates attend an interview. For one candidate, attendance can be modeled as a Bernoulli event with $p = 0.80$. Across 10 independently scheduled candidates under the same conditions, the number who attend follows a binomial model.

$$\text{Expected successes} = n * p = 10 * 0.80 = 8$$

Why independence matters

The binomial model assumes trials are independent and have the same success probability. If one outage causes several candidates to miss interviews together, those attendance outcomes are not independent. The mathematics may still produce a number, but the model no longer matches reality.

Quick Revision Notes

- Probability is between 0 and 1.
- Conditional probability answers questions after information is known.
- Independence means one event does not change the probability of another.
- Bernoulli models one yes/no trial.
- Binomial models the count of successes across repeated Bernoulli trials.

Practice Questions

- Q1** What is the complement of an event with probability 0.72?
- Q2** Why is conditional probability useful in funnels?
- Q3** Give one example of dependent events.
- Q4** What assumptions define a binomial setting?
- Q5** If $n=50$ and $p=0.2$, what is the expected number of successes?

Hands-on Exercises

7. Build a two-stage funnel and calculate conditional conversion.
8. Identify a process at work that violates independence.
9. Model a yes/no outcome as a Bernoulli variable and explain p .

04

Sampling, Standard Error, and the Central Limit Theorem

The sample tells you what happened in the data you observed. Inference asks what that sample suggests about the larger population.

Learning Objectives

- Distinguish population, sample, parameter, and statistic
- Recognize common sampling bias
- Understand sampling variability
- Interpret standard error
- Explain the practical meaning of the Central Limit Theorem

Population vs sample

A population is the full group you want to understand. A sample is the subset you actually observe. A parameter describes the population; a statistic describes the sample. Inference uses sample statistics to learn about unknown population parameters.

Term	Example
Population	all active users this quarter
Sample	2,000 users selected for analysis
Parameter	true average session time of all active users
Statistic	average session time in the 2,000-user sample

Standard error

If you repeatedly drew samples from the same population, the sample mean would change from sample to sample. Standard error summarizes that expected sampling variability.

Standard error of the mean

$$SE = \text{sample SD} / \sqrt{n}$$

Larger samples produce smaller standard errors, all else equal.

COMMON MISTAKE

A large sample reduces random sampling error, but it does not fix biased sampling. Ten million unrepresentative observations can still produce a very precise estimate of the wrong quantity.

Central Limit Theorem

Under broad conditions, the distribution of sample means becomes approximately normal as sample size grows, even when the individual observations are not normally distributed. This is one reason confidence intervals and hypothesis tests work so widely in practice.

Professional example

Suppose a product team estimates average onboarding time from a sample of 400 users. If the sample SD is 8 minutes, $SE = 8 / \sqrt{400} = 0.4$ minutes. That SE measures uncertainty in the estimated mean, not the variability between individual users.

Example

$$SE = 8 / \sqrt{400} = 0.4 \text{ minutes}$$

Quick Revision Notes

- Population is the target group; sample is the observed subset.
- A statistic estimates a population parameter.
- Sampling bias is different from random sampling variability.
- Standard error measures uncertainty of an estimator.
- SE usually decreases as sample size increases.

Practice Questions

- Q1** What is the difference between SD and SE?
- Q2** Does a larger sample eliminate selection bias? Why not?
- Q3** What does the Central Limit Theorem say about sample means?
- Q4** If SD=12 and n=144, what is SE?
- Q5** Give one example of a biased sample.

Hands-on Exercises

10. Take repeated random samples from a dataset and compare their means.
11. Calculate SE for the same SD at n=25, 100, and 400.
12. Write a one-sentence sampling limitation for a dashboard metric.

05

Confidence Intervals and Estimation

A point estimate says "our best guess." An interval says "and here is how uncertain that guess is."

Learning Objectives

- Interpret confidence intervals correctly
- Construct an approximate interval for a mean or proportion
- Understand margin of error
- Explain how sample size affects interval width
- Separate statistical precision from practical importance

Main Concepts

A confidence interval gives a range of parameter values that are reasonably compatible with the observed sample under the method used. A 95% confidence procedure is designed so that, over repeated samples, about 95% of the intervals produced would contain the true parameter.

$$\text{Approximate 95\% interval} \\ \text{estimate} \pm 1.96 * SE$$

The 1.96 multiplier comes from the standard normal distribution.

COMMON MISTAKE

A 95% confidence interval does not mean there is a 95% probability that the fixed true parameter lies inside this particular interval. The 95% describes the long-run performance of the procedure.

Worked Example - offer acceptance rate

Suppose 240 of 400 offers were accepted. The sample acceptance rate is 0.60. The estimated standard error for a proportion is about $\sqrt{p(1-p)/n} = \sqrt{0.60 \cdot 0.40/400} = 0.0245$.

$$\text{95\% CI} \\ 0.60 \pm 1.96 * 0.0245 = \text{approximately } 0.552 \text{ to } 0.648$$

A practical summary: acceptance is estimated at 60%, with a 95% interval of about 55% to 65%.

Margin of error and sample size

Larger samples narrow confidence intervals because they reduce standard error. But beyond a point, very large samples can make tiny, practically irrelevant differences look extremely precise. Precision should support a decision, not replace judgment.

CAREERSENSE INSIGHT

When comparing interview success rates across roles, report both the rate and its interval. A 70% rate from 20 interviews is far less certain than a 70% rate from 2,000 interviews.

Quick Revision Notes

- Confidence intervals communicate uncertainty around estimates.
- Approximate 95% intervals often use estimate $\pm 1.96 \times SE$.
- Larger samples usually create narrower intervals.
- Wide intervals signal limited precision.
- Precision is not the same as business importance.

Practice Questions

- Q1** What does a 95% confidence procedure guarantee in repeated sampling?
- Q2** What happens to interval width as n grows?
- Q3** Why can the same point estimate have different uncertainty?
- Q4** Interpret a conversion estimate of 12% with CI 9%-15%.
- Q5** What is the difference between precision and practical importance?

Hands-on Exercises

13. Construct an approximate 95% interval for a sample mean.
14. Compare intervals for the same proportion at $n=100$ and $n=1,000$.
15. Rewrite a dashboard point estimate as an estimate plus uncertainty statement.

06

Hypothesis Testing and p-values

A hypothesis test asks whether the data are surprising under a specific null model. It does not tell you whether a result matters to the business.

Learning Objectives

- Define null and alternative hypotheses
- Interpret a p-value correctly
- Understand Type I and Type II errors
- Distinguish statistical significance from effect size
- Recognize the role of test assumptions and multiple comparisons

Null and alternative hypotheses

The null hypothesis usually represents no effect, no difference, or a reference value. The alternative represents the effect or difference the analysis is designed to detect. The test statistic measures how far the observed data are from what the null model predicts.

Probability, assuming the null model, of observing a result at least as extreme as the one obtained

Decision reality	Meaning
Type I error	reject a true null - false positive
Type II error	fail to reject a false null - false negative
Power	probability of detecting an effect that is truly present

COMMON MISTAKE

A p-value is not the probability that the null hypothesis is true. It is calculated assuming the null model and describes the extremeness of the observed data under that assumption.

Worked Example - new interview preparation flow

A team compares interview pass rates between the current flow and a redesigned flow. Suppose the difference is +4 percentage points with $p = 0.03$. At a conventional 0.05 threshold, the result is statistically significant - but the team still needs to decide whether a 4-point lift is large enough to justify rollout cost and operational risk.

Effect size first

Statistical significance depends on both effect size and sample size. With enough observations, tiny differences can produce small p-values. Professional analysis reports the magnitude of the effect, an interval, and the p-value - in that order of importance for decision-making.

BEST PRACTICE

Before running the test, define the primary metric, minimum meaningful effect, analysis population, and stopping rule. This reduces the temptation to keep testing until something becomes significant.

Quick Revision Notes

- Null hypothesis defines the reference model.
- p-values measure extremeness under the null, not the probability the null is true.
- Type I error is a false positive; Type II is a false negative.
- Power is the chance of detecting a real effect.
- Effect size and uncertainty matter more than a significance label alone.

Practice Questions

- Q1** What does $p=0.04$ mean?
- Q2** What is a Type I error?
- Q3** How does larger n affect the ability to detect small effects?
- Q4** Why should effect size be reported with p-values?
- Q5** What risk comes from testing many metrics?

Hands-on Exercises

16. Write null and alternative hypotheses for a conversion-rate comparison.
17. Choose a minimum effect that would matter for a real business metric.
18. Take a reported significant result and rewrite the conclusion without the phrase "proved."

07

Correlation and Linear Regression

Two variables moving together does not automatically mean one causes the other. Regression is a model of association, and its usefulness depends on assumptions and context.

Learning Objectives

- Interpret correlation direction and strength
- Understand covariance vs correlation
- Fit and interpret a simple linear regression conceptually
- Interpret slope, intercept, and R-squared
- Recognize confounding, nonlinearity, and extrapolation risks

Correlation

Pearson correlation ranges from -1 to +1 and measures the strength of a linear relationship. A value near zero means little linear association, not necessarily no relationship. A curved relationship can have correlation near zero while still being highly predictable.

Correlation

$$r = \text{standardized covariance of X and Y}$$

Unitless and bounded between -1 and +1.

COMMON MISTAKE

High correlation does not establish causation. Both variables may be driven by a third factor, the direction may be reversed, or the relationship may be coincidental.

Simple linear regression

Model

$$Y = \text{intercept} + \text{slope} * X + \text{error}$$

The slope estimates the average change in Y associated with a one-unit increase in X, within the model. The intercept is the predicted Y when X=0, which may or may not be meaningful. R-squared describes the proportion of variance in Y explained by the model in-sample.

Worked Example - practice sessions and interview score

Suppose a regression estimates: $\text{interview_score} = 62 + 2.4 * \text{practice_sessions}$. The slope of 2.4 says each additional practice session is associated with an average 2.4-point higher interview score in the observed data. It does not prove the session caused the improvement; more motivated candidates may both practise more and perform better.

BEST PRACTICE

Plot the data before fitting the model. A regression line can hide outliers, clusters, changing variance, and curved relationships that make a single slope misleading.

Extrapolation

A model trained on 0-10 practice sessions should not be trusted automatically at 50 sessions. Regression describes the relationship in the observed range; predictions far outside that range rely on assumptions the data did not test.

Quick Revision Notes

- Correlation measures linear association.
- Correlation does not imply causation.
- Regression slope is an average conditional association.
- R-squared is not a measure of causal validity.
- Extrapolation beyond observed data is risky.

Practice Questions

- Q1** What does $r = -0.8$ suggest?
- Q2** Can $r = 0$ occur with a strong nonlinear relationship?
- Q3** Interpret a regression slope in plain language.
- Q4** What does R-squared summarize?
- Q5** Why can confounding create misleading regression results?

Hands-on Exercises

19. Create a scatterplot in your preferred tool and describe shape before calculating correlation.
20. Write a regression equation and explain each term.
21. List three potential confounders for a relationship you care about.

08

Experiments and A/B Testing

A controlled experiment is powerful because random assignment can make groups comparable before the treatment occurs.

Learning Objectives

- Identify treatment, control, outcome, and unit of randomization
- Explain why randomization supports causal inference
- Choose primary and guardrail metrics
- Understand sample-size and power trade-offs
- Avoid peeking, contamination, and post-hoc metric selection

Experiment anatomy

Element	Example
Unit	candidate account
Treatment	new AI interview feedback flow
Control	current feedback flow
Primary metric	next-interview pass rate
Guardrail	session completion rate
Randomization	50/50 assignment by candidate

Randomization balances both observed and unobserved factors on average. That is what makes the difference in outcomes interpretable as a treatment effect, assuming the experiment is implemented cleanly.

BEST PRACTICE

Choose the primary metric and minimum meaningful effect before launch. If every metric is primary after the results arrive, the experiment has no disciplined decision rule.

Absolute vs relative lift

If conversion rises from 10% to 12%, the absolute lift is 2 percentage points. The relative lift is 20%. Both are correct, but they answer different questions. Always label which one you report.

Lift

$$\text{absolute} = 12\% - 10\% = 2 \text{ points; relative} = 2/10 = 20\%$$

Common experiment failures

- Sample ratio mismatch: the observed allocation differs materially from the intended randomization.
- Contamination: users in control are exposed to treatment features.
- Peeking: repeatedly checking significance and stopping at the first favorable result.
- Novelty effects: short-term behavior changes because the experience is new.
- Metric movement without user value: a primary metric improves while a guardrail worsens.

Decision framing

A strong experiment conclusion states the estimated lift, confidence interval, practical threshold, guardrails, and any implementation limitations. The question is not "was $p < 0.05$?" but "does the total evidence justify the decision?"

Quick Revision Notes

- Randomization is the foundation of causal A/B testing.
- Primary metrics should be pre-specified.
- Guardrails detect harmful side effects.
- Absolute and relative lift must be labeled clearly.
- Peeking and multiple metrics inflate false-positive risk.

Practice Questions

- Q1** Why does randomization help causal inference?
- Q2** What is a guardrail metric?
- Q3** Distinguish absolute and relative lift.
- Q4** What is contamination?
- Q5** Why is peeking a statistical problem?

Hands-on Exercises

22. Design an A/B test for a career platform feature.
23. Choose one primary metric and two guardrails.
24. Write a launch decision rule before seeing results.

09

Statistical Pitfalls, Bias, and Causality

The most dangerous statistical mistakes often happen before the calculation: who entered the dataset, how the metric was defined, and what comparisons were allowed.

Learning Objectives

- Recognize selection, survivorship, measurement, and confirmation bias
- Understand confounding and Simpson's paradox
- Identify data leakage and post-treatment bias
- Separate prediction from causal inference
- Communicate limitations without making analysis useless

Bias patterns professionals should recognize

Bias / pitfall	Example
Selection bias	survey respondents differ from non-respondents
Survivorship bias	studying only candidates who completed the process
Measurement bias	different teams record the same event differently
Confounding	experience affects both training uptake and interview performance
Data leakage	using future information to predict an earlier outcome

Simpson's paradox

An overall trend can reverse after data are separated into meaningful groups. For example, one interview-prep program might appear to have a lower pass rate overall because it is used more often by candidates applying to harder roles. Within each role level, it could actually perform better.

PROFESSIONAL HABIT

Whenever a comparison looks surprising, ask which variables determine who enters each group. Segment before concluding.

Prediction is not causation

A model can predict who is likely to receive an offer without telling you which intervention would increase that candidate's chance of receiving one. Prediction asks what will happen; causal inference asks what would happen under a different action.

Communicating limitations

Avoid two extremes: pretending the analysis is certain, or listing so many caveats that no decision can be made. State the key limitation, explain the direction of likely bias if known, and say what evidence would reduce uncertainty next.

EXAMPLE LIMITATION

"Candidates self-selected into the coaching program, so the observed 9-point pass-rate difference should be treated as an association rather than a causal effect. A randomized rollout would provide stronger evidence."

Quick Revision Notes

- Bias is a systematic distortion, not random noise.
- Confounding can create misleading associations.
- Simpson's paradox can reverse aggregate trends.
- Data leakage makes predictive performance look better than it will be in reality.
- Predictive accuracy does not establish intervention impact.

Practice Questions

- Q1** Give an example of selection bias.
- Q2** What is confounding?
- Q3** How can an aggregate trend reverse after segmentation?
- Q4** Why is data leakage dangerous?
- Q5** How does a causal question differ from a predictive question?

Hands-on Exercises

- 25. Audit one dataset for survivorship bias.
- 26. Write a causal and a predictive version of the same business question.
- 27. Take a surprising aggregate metric and identify two useful segmentation variables.

10

Professional Case Study: CareerSense Job-Search Analytics

Bring the full toolkit together: define metrics, describe the funnel, quantify uncertainty, compare groups, and make a decision without overstating causality.

Learning Objectives

- Design a statistically coherent job-search analytics dataset
- Build funnel metrics with correct denominators
- Compare cohorts with intervals and effect sizes
- Design an experiment for a CareerSense feature
- Produce an executive-ready statistical narrative

Scenario

CareerSense wants to understand whether candidates who complete at least three AI interview-practice sessions are more likely to reach an offer. The platform has application, interview, practice, and offer events for 8,000 candidates.

Step 1 - define the grain

Use one row per candidate for the primary cohort analysis. Candidate-level features include target role, years of experience, number of applications, number of practice sessions, interview count, and whether at least one offer was received.

Step 2 - descriptive comparison

Group	Candidates	Offer rate	Median applications
0-2 practice sessions	5,200	11.8%	22
3+ practice sessions	2,800	18.6%	25

The raw difference is +6.8 percentage points. That is an association, not yet a causal effect, because practice usage is self-selected.

$$\text{Raw effect size} \\ 18.6\% - 11.8\% = +6.8 \text{ percentage points}$$

Step 3 - quantify uncertainty

Report confidence intervals for each rate and for the difference. Large samples may make the difference very precise, but precision does not remove confounding. Role difficulty, motivation, prior experience, and application volume may differ between groups.

Step 4 - model responsibly

A logistic regression could adjust for measured variables such as experience, target role, geography, and application volume. If the practice-session coefficient remains positive, that strengthens the association but still does not fully establish causality because unmeasured confounding can remain.

Step 5 - design the causal test

Randomly offer an enhanced interview-practice experience to eligible candidates and compare a pre-specified outcome such as interview-to-offer conversion. Use completion and user satisfaction as guardrails. Pre-register the minimum meaningful lift and analysis window.

EXECUTIVE CONCLUSION

"Candidates completing 3+ practice sessions had a 6.8-point higher observed offer rate. The association remains after adjustment for major observed factors, but self-selection limits causal interpretation. We recommend a randomized rollout to estimate the true incremental impact."

Mini project deliverable

- One-page metric definition sheet
- Candidate-level analytical dataset
- Descriptive funnel table with uncertainty
- Adjusted regression summary
- A/B test design with primary metric and guardrails
- Five-sentence executive narrative

Quick Revision Notes

- Start with grain and denominators.
- Report effect size before significance.
- Intervals quantify precision, not causal validity.
- Regression can adjust measured confounders but not guarantee causality.
- Randomized experiments provide stronger causal evidence when feasible.

Practice Questions

- Q1** Why is the raw 6.8-point difference not automatically causal?
- Q2** What confounders might affect both practice usage and offer rate?
- Q3** Why might logistic regression help?
- Q4** What would you randomize in a follow-up experiment?
- Q5** What should an executive conclusion include?

Hands-on Exercises

28. Create a candidate-level schema for the case study.
29. Write the exact formula for interview-to-offer conversion.
30. Draft a pre-analysis plan with one primary metric, one minimum effect, and two guardrails.

Course Summary

Statistics for data professionals is not a collection of isolated formulas. It is a reasoning system for moving from observations to decisions while accounting for uncertainty.

You began with data types, grain, center, spread, and distributions. You then moved into probability and sampling, where statistics become inferential: a sample is used to learn about a larger population. Standard error and confidence intervals gave you a language for precision. Hypothesis testing introduced evidence against a reference model, while emphasizing that effect size and practical importance matter more than a significance label.

The later chapters focused on relationships and decisions: correlation and regression for association, controlled experiments for causal effects, and the bias patterns that can undermine both. The final case study connected these ideas into a professional workflow: define the dataset, calculate descriptive metrics, quantify uncertainty, adjust cautiously, and design stronger evidence when needed.

THE PROFESSIONAL STANDARD

A strong analyst can answer five questions: What exactly was measured? How large is the effect? How uncertain is the estimate? What assumptions support the conclusion? What decision does the evidence justify?

Quick Revision Notes - The Whole Book

Descriptive statistics

- Define grain before metrics.
- Mean is sensitive to extremes; median and IQR are robust.
- Distribution shape determines which summaries are useful.

Probability and inference

- Probability models uncertainty.
- Standard error measures sampling variability of an estimator.
- Confidence intervals communicate precision around estimates.

Testing and modeling

- p-values describe extremeness under the null; they are not the probability the null is true.
- Report effect size and interval alongside significance.
- Correlation and regression describe association unless stronger causal assumptions are justified.

Experiments and judgment

- Randomization supports causal inference.
- Pre-specify metrics and decision rules.
- Bias, confounding, leakage, and selection can invalidate otherwise correct calculations.

REFERENCE

Professional Glossary

Term	Meaning in plain English
Population	The full group the analysis aims to understand.
Sample	The observed subset used to estimate population properties.
Parameter	An unknown numerical property of a population.
Statistic	A numerical summary calculated from a sample.
Mean	Arithmetic average.
Median	Middle value after sorting.
Variance	Average squared deviation from the mean.
Standard deviation	Square root of variance; spread in original units.
IQR	$Q3 - Q1$; spread of the middle 50%.
Percentile	Value below which a stated percentage of observations fall.
z-score	Standardized distance from the mean in SD units.
Probability	Number from 0 to 1 representing uncertainty under a model.
Conditional probability	Probability of one event given another is known.
Standard error	Sampling variability of an estimator.
Central Limit Theorem	Result explaining why many sampling distributions become approximately normal.
Confidence interval	Range produced by a procedure designed to capture the parameter at a stated long-run rate.
Null hypothesis	Reference model used in a hypothesis test.
p-value	Probability under the null of data at least as extreme as observed.
Type I error	False positive: rejecting a true null.
Type II error	False negative: failing to reject a false null.
Power	Probability of detecting a real effect under a specified alternative.
Effect size	Magnitude of a difference or relationship.
Correlation	Standardized measure of linear association.
Regression	Model relating an outcome to one or more predictors.

R-squared	Share of outcome variance explained by a regression model in-sample.
Randomization	Assignment by chance used to create comparable experimental groups.
Confounding	A third factor related to both exposure and outcome that can distort association.
Selection bias	Systematic difference between observed data and target population.
Data leakage	Use of information unavailable at prediction time, inflating model performance.

FINAL CHECK

20-Question Professional Quiz

01. When is the median usually preferable to the mean?
02. What does standard deviation measure?
03. What does the 90th percentile represent?
04. What does $z=2$ mean?
05. What is conditional probability?
06. What assumption distinguishes independent events?
07. What is the difference between a population parameter and a sample statistic?
08. What does standard error measure?
09. What practical idea does the Central Limit Theorem support?
10. What does a 95% confidence procedure mean in repeated sampling?
11. What does a p-value of 0.03 mean?
12. What is a Type I error?
13. Why should effect size be reported with significance?
14. What does correlation measure?
15. Why does correlation not establish causation?
16. What does a regression slope represent?
17. Why is random assignment valuable in an A/B test?
18. What is a guardrail metric?
19. What is selection bias?
20. Why can regression adjustment fail to prove causality?

Answer Key

01. When the distribution is skewed or contains extreme values and the typical case matters.
02. The spread of observations around the mean, in the variable's original units.
03. The value at or below which 90% of observations fall.
04. The observation is two standard deviations above the mean.
05. The probability of event A after learning that event B occurred.
06. Knowing one event occurs does not change the probability of the other.
07. A parameter describes the population; a statistic is computed from the sample.
08. The expected sampling variability of an estimator such as the sample mean.
09. It helps justify approximately normal sampling distributions for many estimators at sufficient sample size.
10. Across repeated samples, about 95% of intervals produced by the procedure would contain the true parameter.
11. Assuming the null model, results at least this extreme would occur about 3% of the time.
12. Rejecting a null hypothesis that is actually true.
13. Because statistical significance does not tell you whether the magnitude is useful or important.
14. The strength and direction of a linear association.
15. Association may be produced by confounding, reverse direction, or coincidence.
16. The average modeled change in Y associated with a one-unit increase in X, holding other included predictors fixed.
17. It makes treatment and control comparable on average, supporting causal interpretation.
18. A metric used to ensure treatment does not cause unacceptable harm elsewhere.
19. A systematic difference between observed units and the population or comparison you intend to represent.
20. Adjustment only controls measured variables and depends on model assumptions; unmeasured confounding can remain.



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